System of Linear equations:

A system of linear equation is a list of linear equations with the same unknowns. In particular, a system of m linear equations are

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\cdots\cdots\cdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

Short form, $\sum_{j=1}^{n} a_{ij}x_j = b_i \quad (i=1,2,\ldots,m)$.

matrix form,

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}_{m\times n}}_{\text{coefficient matrix}} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}_{n\times 1} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}_{m\times 1}$$

Non-homogeneous system of linear equation:

Let $\sum_{j=1}^{n} a_{ij}x_j = b_i \ (i=1,2,\ldots,m)$ be a system of m linear equations in the n unknowns $x_1, x_2, \ldots, x_n$ where $a_{ij}, b_i \in \mathbb{R}$.

If atleast one constant of the set $\{b_1, b_2, \ldots, b_m\}$ is not zero then the system is called a non-homogeneous system of linear equation.

Homogeneous System of linear equation:

The system $\sum_{j=1}^{n} a_{ij} x_j = 0$ $(i = 1, 2, \ldots m)$ is called a homogeneous system of m linear equations where $a_{ij} \in \mathbb{R}$.

It has a sol^n since for the case $(0, 0, \ldots, 0) \in \mathbb{R}$ is a solution. this solution is called the zero or trivial solution of this system.

If $(k_1, k_2, \ldots, k_n) \in \mathbb{R}$ is a solution of the homogeneous system if at least one k_i is not zero, then it is called a non-zero or non-trivial solution of the homogeneous system.

* A homogeneous system has a non-trivial sol^n if the number of equation is less than the number of unknown.

Particular and General solution:

Let $\sum_{j=1}^{n} a_{ij} x_j = b_i$ $(i=1,2,\dots,n)$ be a (1) system of m linear equations which may be homogeneous or non-homogeneous.

If $(k_1, k_2, \dots, k_n) \in \mathbb{R}$ is a solution of the system (1) then all the value is called a particular solution of the system (1).

The set of all particular solutions of the system (1) is called a General soln or soln set of the system.

If the system is homogeneous then the General solution is called the homogeneous space.

consistent and in-consistent system of linear equation:

A system of linear equations $\sum_{j=1}^{n} a_{ij} x_j = b_i$ $(i=1,2,\dots,m)$ is said to be consistent if it has unique solution or more than one solution.

Every homogeneous system is consistent.

(ii) consistent system of linear equation will have unique solution if in its echelon form free variables does not exist.

(ii) consistent system of linear equation will have more than one solution if in its echelon form free variable exist.

⇒ A system of linear equations $\sum_{j=1}^{n} a_{ij} x_j = b_i$ is said to be ~~consistent~~ inconsistent if it has no solution.

(iii) Inconsistent system of linear equation will have no solution if in its echelon form at least one equation will appear of the form $0 = b$ [where $b \neq 0$]

Degenerate linear equation: A linear equation is said to be degenerate if all the coefficient are zero that is if it has the form

$$0 \cdot x_1 + 0 \cdot x_2 + \cdots + 0 \cdot x_n = b$$

If $b \neq 0$, then the equation has no solution.
If $b = 0$, every vector $u = (k_1, k_2, \ldots, k_n)$ in K^n is a solution.